

Submit your Jupyter notebook with the answers to gradescope by 1:50pm.

1. Define a piecewise function $f(x)$, defined as follows:

$$f(x) = \begin{cases} -x & \text{if } x < -1, \\ 0 & \text{if } x \in [-1, 1], \\ +x & \text{if } x > 1. \end{cases}$$

Verify that the derivative of your function is also piecewise.

Make a plot of f over $[-2, 2]$ to show your definition is correct.

2. Consider the recursive function

$$f(n, 0) = 1, \quad f(0, m) = 1, \quad f(n, m) = 2n f(n-1, m) + 3m f(n, m-1).$$

Define a Python function `F` which returns $f(n, m)$, using a dictionary for memoization. Make sure your `F` can compute $f(100, 100)$.